

Rainfall Estimation

This note summarizes results for a collection of 100 patches. All aggregate figures and tables that follow should therefore be read as patch-level summaries over this hundred-patch evaluation set, rather than as statistics from a single patch or from the full archive of available patches.

Relative Absolute Error Rainy-Pixel Distance Profile

For a patch P , let $\Omega(P)$ denote its pixel set, let $R_P : \Omega(P) \rightarrow \mathbb{R}_{\geq 0}$ be the ground-truth rainfall field, and let $\hat{R}_{P,s} : \Omega(P) \rightarrow \mathbb{R}_{\geq 0}$ be the rainfall field predicted by solver s . We define a pixel p to be rainy if $R_P(p) \geq 0.6$ mm/h, and we write

$$\Omega_{\text{rain}}(P) = \{p \in \Omega(P) : R_P(p) \geq 0.6\}.$$

For a rainy pixel p , the relative absolute error with respect to solver s is

$$\text{RAE}_{P,s}(p) \triangleq \frac{|R_P(p) - \hat{R}_{P,s}(p)|}{R_P(p)}.$$

To study how prediction quality changes with link proximity, we group pixels by distance to the third-closest link. Let $\mathcal{L}(P)$ denote the set of links contained in patch P . For a pixel $p \in \Omega(P)$, let $c(p) \in \mathbb{R}^2$ denote the center of pixel p . For a link $\ell \in \mathcal{L}(P)$, viewed geometrically as a line segment, let

$$\text{dist}(p, \ell) \triangleq \text{dist}(c(p), \ell)$$

denote the Euclidean distance from the pixel center to that link segment. If the links in patch P are written as $\ell_1, \dots, \ell_{|\mathcal{L}(P)|}$, we define

$$d_3(p) \triangleq \text{the third-order statistic of } \{\text{dist}(p, \ell_j)\}_{j=1}^{|\mathcal{L}(P)|},$$

that is, the distance from the center of pixel p to its third-closest link segment.

For each index i , let b_{i-1} and b_i denote the lower and upper distance cutoffs of the i th bin, and write $B_i = (b_{i-1}, b_i]$. A pixel p belongs to bin B_i exactly when $d_3(p) \in B_i$.

For each patch P , solver s , and distance bin B_i , we then collect the rainy-pixel RAE values in that bin and summarize them by the patch-level median

$$m_{s,i}^{\text{rain}}(P) \triangleq \text{median}\{\text{RAE}_{P,s}(p) : p \in \Omega_{\text{rain}}(P), d_3(p) \in B_i\}.$$

In Figure 1, these patch-level medians are compared across the full benchmark collection $\mathcal{P}$ of 100 patches. There, the displayed distance bins are

$$[0, 125] \text{ m}, (125, 375] \text{ m}, (375, 750] \text{ m}, (750, 1500] \text{ m}, (1500, 3125] \text{ m}, (3125, 6000] \text{ m},$$

with the sparsely populated tail beyond 6000 m merged into the last displayed bin. For fixed solver s and distance bin B_i , let

$$\mu_{s,i}^{\text{rain}} \triangleq \{m_{s,i}^{\text{rain}}(P)\}_{P \in \mathcal{P}}$$

denote the sequence of patch-level rainy-pixel median RAE values across patches. Each box-and-whisker then summarizes this sequence: the lower whisker tip is the minimum of $\mu_{s,i}^{\text{rain}}$, the lower edge of the box is its 25th percentile, the horizontal line segment with the filled circle at its center marks the median, the upper edge of the box is its 75th percentile, and the upper whisker tip is its maximum.

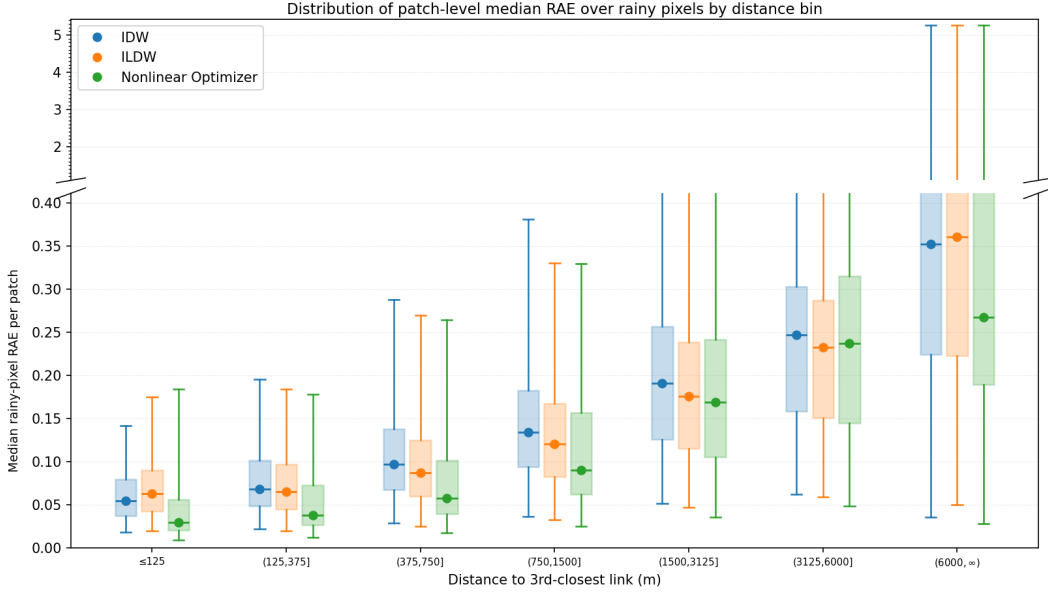


Figure 1: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median rainy-pixel relative-absolute-error (RAE) value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a rainy pixel is a pixel whose ground-truth rainfall is at least the configured wet threshold (set to 0.6 mm/h), and the relative absolute error for a rainy pixel p by solver s is defined as $|R_P(p) - \hat{R}_{P,s}(p)|/R_P(p)$. The corresponding rainy-pixel counts per bin are reported in Table 1.

Absolute Error Non-Rainy-Pixel Distance Profile

For the dry-pixel analysis, we focus on pixels whose ground-truth rainfall falls below the wet threshold. We therefore define the non-rainy pixel set by

$$\Omega_{\text{nonrain}}(P) = \{p \in \Omega(P) : R_P(p) < 0.6\}.$$

For these pixels, relative normalization by $R_P(p)$ is not appropriate because the denominator may be zero or very small. We therefore measure prediction quality by absolute error rather than relative absolute error:

$$\text{AE}_{P,s}(p) = |R_P(p) - \hat{R}_{P,s}(p)|.$$

Distance to third closest link (m)	Wet mean count (% of total)	SD
[0, 125]	4267.0 (1.79%)	960.4
(125, 375]	16737.7 (7.01%)	3803.2
(375, 750]	26809.9 (11.22%)	6707.0
(750, 1500]	51988.5 (21.76%)	14 468.0
(1500, 3125]	88839.4 (37.18%)	27 531.1
(3125, 6000]	45929.4 (19.22%)	15 679.4
(6000, 9000]	3835.8 (1.61%)	2725.7
(9000, ∞)	514.4 (0.22%)	966.9
Total	238922.1 (100%)	

Table 1: Rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 1. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the rainy-pixel count in each bin across patches. The last bin, $(9000, \infty)$, is merged with the previous bin in Figure 1 because it is sparsely populated.

Using the same definition of the third-closest-link distance $d_3(p)$ introduced above, and the same distance bins B_i as in the rainy-pixel analysis, we compute for each patch P , solver s , and distance bin B_i

$$m_{s,i}^{\text{nonrain}}(P) \triangleq \text{median}\{\text{AE}_{P,s}(p) : p \in \Omega_{\text{nonrain}}(P), d_3(p) \in B_i\},$$

and, for fixed s and i , the corresponding box-and-whisker plot summarizes the sequence

$$\mu_{s,i}^{\text{nonrain}} \triangleq \{m_{s,i}^{\text{nonrain}}(P)\}_{P \in \mathcal{P}}$$

of patch-level non-rainy-pixel median AE values across patches. Thus the rainy and non-rainy figures use the same distance-bin construction, but they differ both in the pixel set being summarized and in the error metric being aggregated.

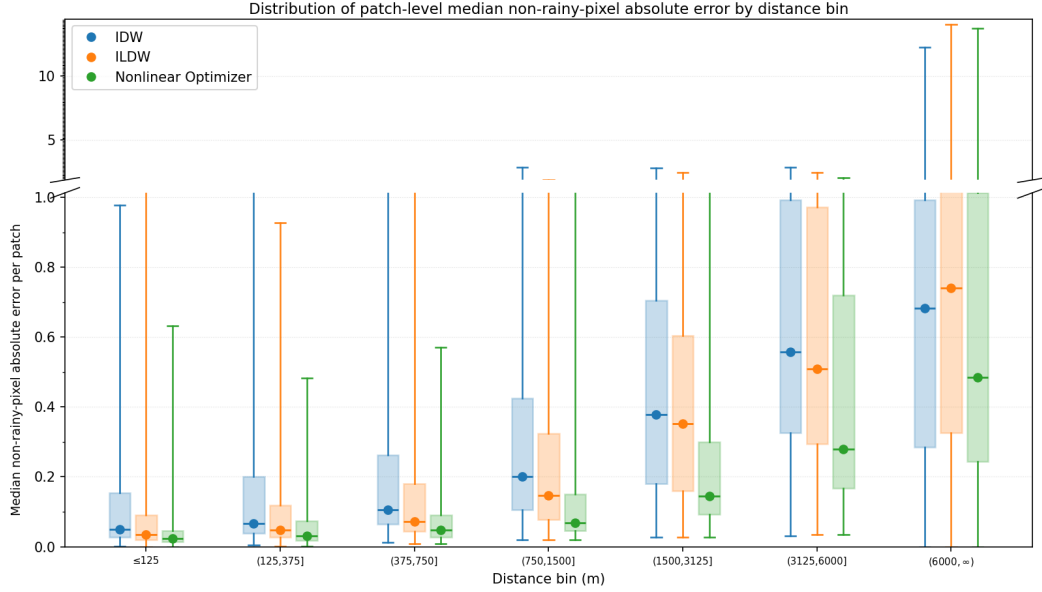


Figure 2: Non-rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median non-rainy-pixel absolute error value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a non-rainy pixel is a pixel whose ground-truth rainfall is below the configured wet threshold (set to 0.6 mm/h), and the absolute error for a non-rainy pixel p by solver s is defined as $|R_P(p) - \hat{R}_{P,s}(p)|$. The corresponding non-rainy-pixel counts per bin are reported in Table 2.

Distance to third closest link (m)	Dry mean count (% of total)	SD
[0, 125]	563.6 (1.84%)	777.6
(125, 375]	2261.1 (7.40%)	3093.5
(375, 750]	3389.1 (11.09%)	4709.5
(750, 1500]	6446.5 (21.09%)	9257.7
(1500, 3125]	10776.2 (35.25%)	15 553.7
(3125, 6000]	6197.2 (20.27%)	8990.0
(6000, 9000]	849.9 (2.78%)	1289.7
(9000, ∞)	85.5 (0.28%)	250.9
Total	30569.2 (100%)	

Table 2: Non-rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 2. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the non-rainy-pixel count in each bin across patches. The last bin, $(9000, \infty)$, is merged with the previous bin in Figure 2 because it is sparsely populated.

Largest-Patch Link Geometry

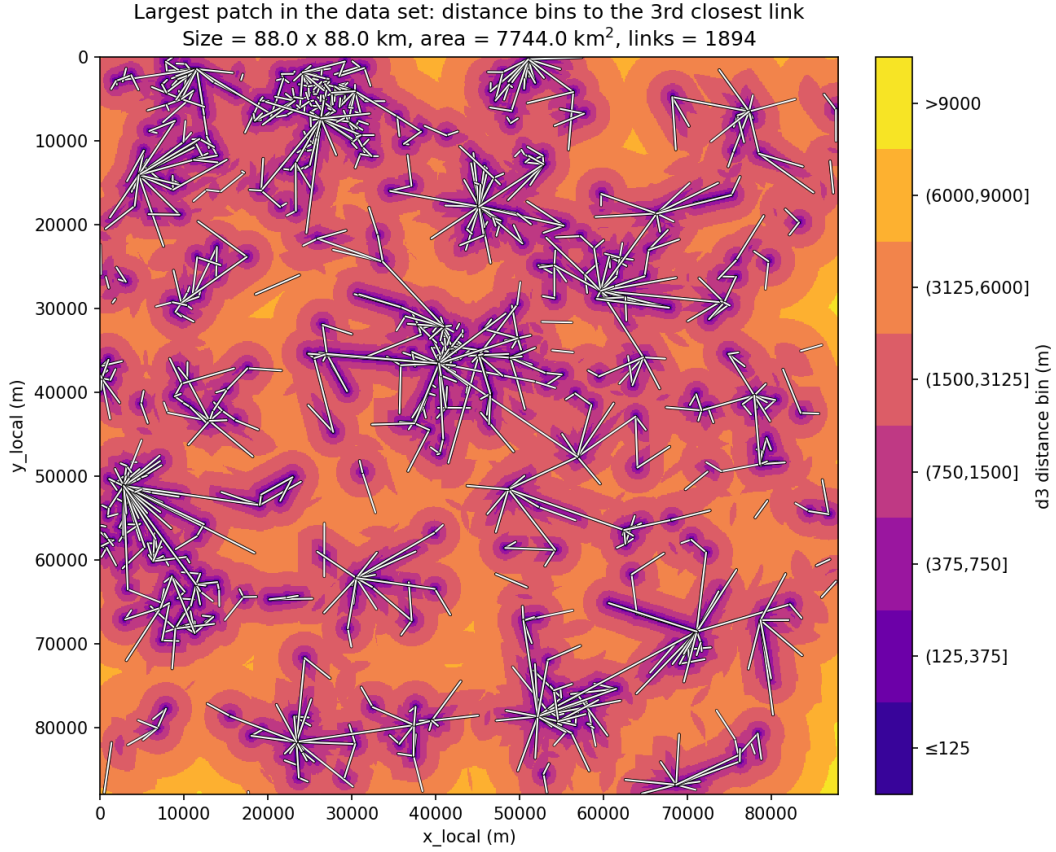


Figure 3: Distance-bin map for the largest evaluated patch in the data set. Each pixel is colored according to the bin induced by its distance to the third closest link, and the complete link geometry is overlaid in white.

The same largest patch is also useful for describing the structure of the underlying link network. In the summaries below, link length refers to the path length of a link, carrier frequency is the microwave-link frequency in GHz, and endpoint degree is the number of links incident to a given endpoint node. Here, an endpoint node is a location at which one or more links meet.

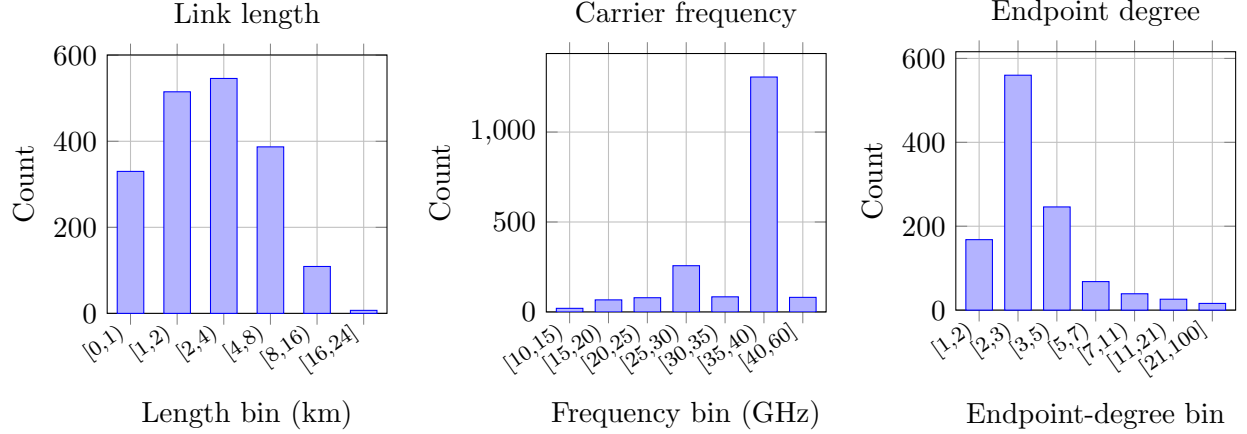


Figure 4: Histogram summaries for the largest patch. Left: link lengths. Middle: carrier frequencies. Right: endpoint-degree histogram. The x-axis gives endpoint degree and the y-axis gives the number of endpoint nodes in each degree bin.

Length bin (km)	Count	%	Frequency bin (GHz)	Count	%	Endpoint-degree bin	Count	%
[0, 1)	330	17.4	[10, 15)	20	1.1	[1, 2)	168	15.0
[1, 2)	515	27.2	[15, 20)	67	3.5	[2, 3)	560	49.9
[2, 4)	546	28.8	[20, 25)	79	4.2	[3, 5)	246	21.9
[4, 8)	387	20.4	[25, 30)	257	13.6	[5, 7)	68	6.1
[8, 16)	109	5.8	[30, 35)	84	4.4	[7, 11)	39	3.5
[16, 24]	7	0.4	[35, 40)	1306	69.0	[11, 21)	26	2.3
			[40, 60]	81	4.3	[21, 100]	16	1.4

Table 3: Tabulated distributions for the largest-patch link network. The first panel counts links by path-length interval, the second counts links by carrier-frequency interval, and the third counts unique endpoint nodes by their graph degree.

Confusion Matrix

IDW			ILDW			Nonlinear Optimizer		
GT	Prediction		GT	Prediction		GT	Prediction	
	Wet	Dry		Wet	Dry		Wet	Dry
Wet	0.893 $\pm$ 0.013	0.002 $\pm$ 0.000	Wet	0.892 $\pm$ 0.013	0.003 $\pm$ 0.000	Wet	0.890 $\pm$ 0.013	0.005 $\pm$ 0.001
Dry	0.040 $\pm$ 0.004	0.065 $\pm$ 0.010	Dry	0.037 $\pm$ 0.004	0.068 $\pm$ 0.010	Dry	0.030 $\pm$ 0.004	0.075 $\pm$ 0.010

Table 4: Patch-averaged wet/dry confusion matrices for IDW, ILDW, and the nonlinear optimizer at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels: top-left is TP, top-right is FN, bottom-left is FP, and bottom-right is TN. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM).

Patch Geometry Overview

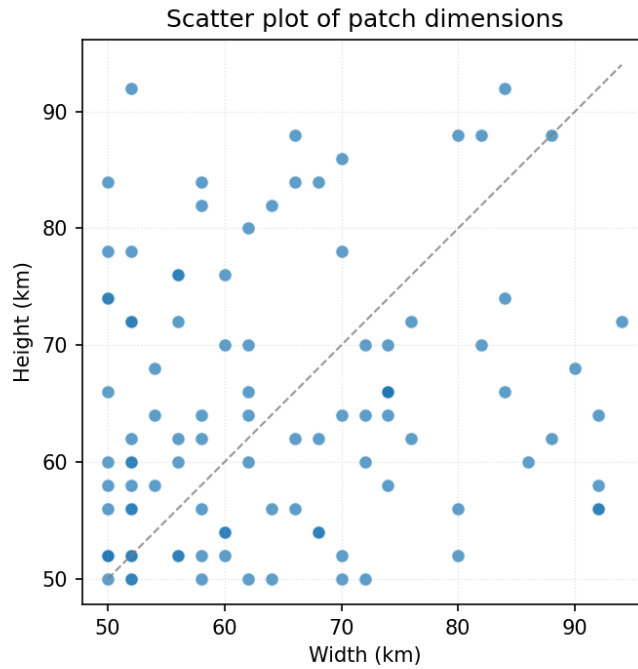


Figure 5: Patch width versus patch height for the 100 patches, with one dot per patch. Across patches, the mean patch area is 269,491 pixels with SD of 76,755 pixels (pixels are squares with side length 125m).

Patch size (km)	# links	1-multiplicity	2-multiplicity
50×50	939	99	420
62×60	1254	128	563
72×70	1507	157	675

Table 5: Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with approximately 10% of links having multiplicity 1.

J_{atten} Summary

Solver	Mean J_{atten} (dB ² /km)	SD
IDW	0.01522	0.03808
ILDW	0.00289	0.00757
Nonlinear Optimizer	0.00020	0.00103

Table 6: Patch-level summary of J_{atten} across the 100 evaluated patches. For a given patch, J_{atten} is the attenuation-fit term $J_{\text{atten}} = \frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{(\hat{A}_{\ell} - A_{\ell}^{\text{obs}})^2}{|\ell|}$, where $\hat{A}_{\ell}$ is the attenuation implied by the estimated rainfall field on link ℓ , A_{ℓ}^{obs} is the observed attenuation, and $|\ell|$ is the link length in kilometers. The table reports the mean and standard deviation of this per-patch quantity for each solver. Lower values indicate better agreement with the observed link attenuations.

Absolute Attenuation Error per km

Solver	Mean (dB/km)	SD
IDW	0.05034	0.03413
ILDW	0.01356	0.00948
Nonlinear Optimizer	0.00230	0.00437

Table 7: Patch-level summary of the mean absolute attenuation error per km. For a given patch, we compute $\frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{|\hat{A}_{\ell} - A_{\ell}^{\text{obs}}|}{|\ell|}$, where $\hat{A}_{\ell}$ is the attenuation induced by the estimated rainfall field on link ℓ , A_{ℓ}^{obs} is the observed attenuation, and $|\ell|$ is the link length in kilometers. Thus, the absolute attenuation error of every link is normalized by the link’s length. The table reports the mean and standard deviation of this quantity across the 100 evaluated patches for each solver. Lower values indicate better agreement with the observed link attenuations.

Qualitative Case Studies

The three case studies below pair a geographic view of the selected patch with the corresponding rainy-pixel solver outputs. The timestamp in each caption identifies the radar scene, and patch 0 denotes the selected patch within that scene. This layout is intended to keep the geographic context and the solver behavior visually tied to one another.

For the nonlinear optimizer, the estimated rainfall field is computed with the L-BFGS-B algorithm, initialized from the ILDW field and constrained to remain nonnegative at every pixel. The objective minimized by the optimizer is a weighted sum of four terms,

$$\alpha_{\text{atten}} J_{\text{atten}} + \alpha_{1d} J_{1d} + \alpha_{2d} J_{2d} + \alpha_{\text{total}} J_{\text{total}},$$

where the instance-specific weights are chosen from the corresponding ILDW objective values, namely

$$\alpha_{\text{atten}} = \frac{1}{J_{\text{atten}}^{\text{ILDW}}}, \quad \alpha_{1d} = \frac{1}{J_{1d}^{\text{ILDW}}}, \quad \alpha_{2d} = \frac{1}{2 J_{2d}^{\text{ILDW}}}, \quad \alpha_{\text{total}} = \frac{1}{J_{\text{total}}^{\text{ILDW}}}.$$

The optimizer-trace plots included below show, for each selected case study, how the weighted objective and its weighted constituent terms decrease over the optimization iterations.

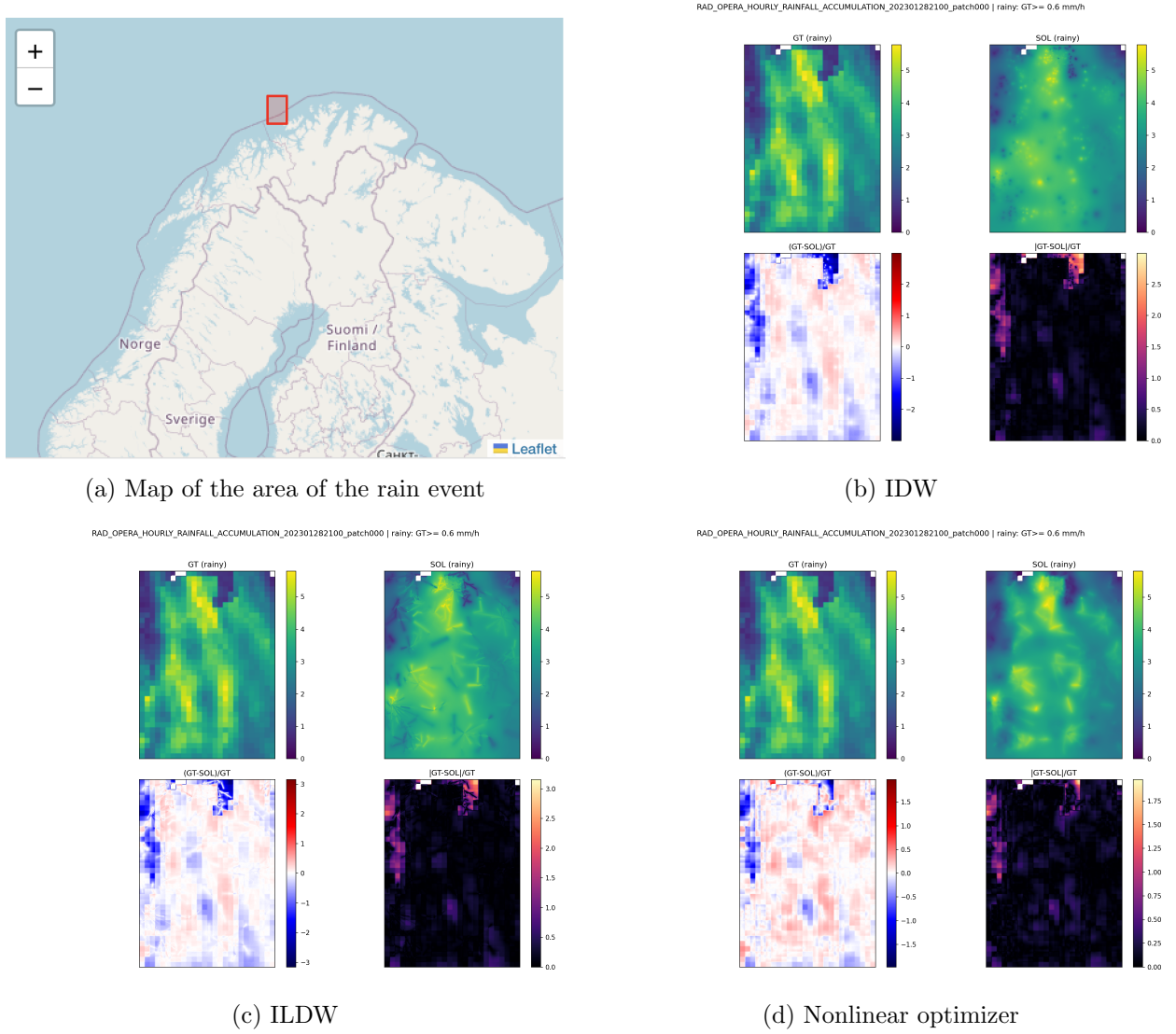


Figure 6: Case study for a rain event registered on January 28th, 2023. The map on the left shows the selected patch footprint in geographic context. The three rainy-pixel solver panels correspond to that same patch and show, for each solver, the ground truth field R_P , the estimate $\hat{R}_{P,s}$, the signed relative error $(R_P - \hat{R}_{P,s})/R_P$, and the absolute relative error $|R_P - \hat{R}_{P,s}|/R_P$. This layout keeps the location of the event visually tied to the reconstruction quality of each solver.

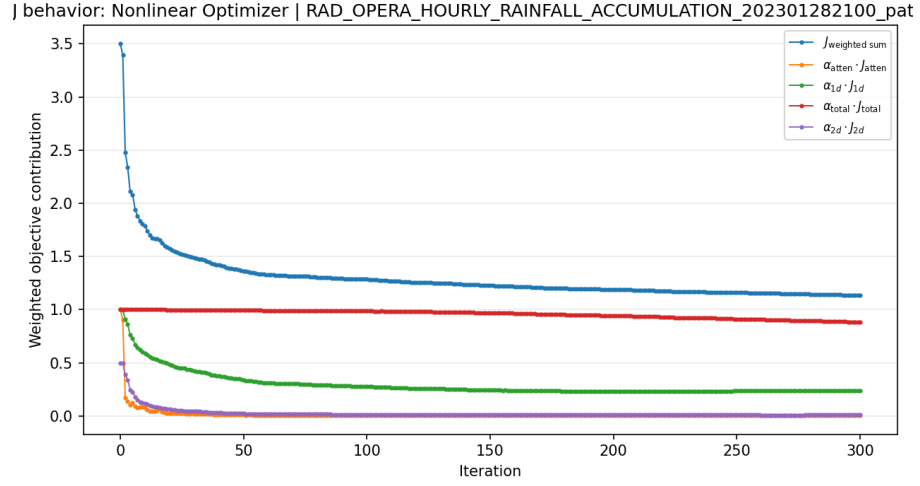
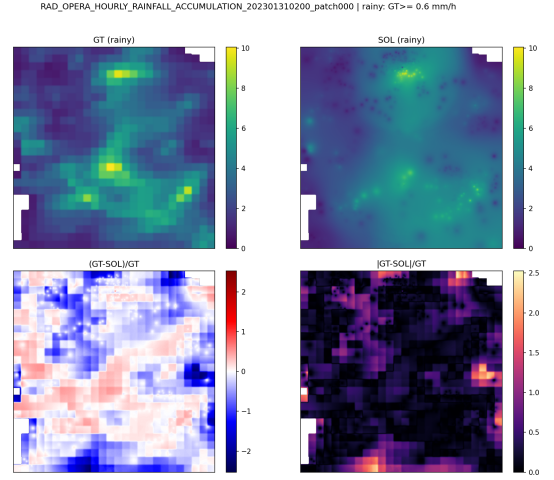


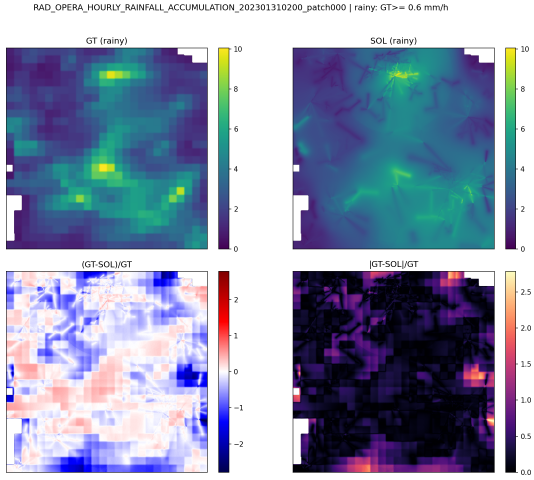
Figure 7: Optimizer trace for the January 28th, 2023 case study. The plot shows the reduction of the weighted objective $J_{\text{weighted sum}}$ and of the weighted constituent terms over L-BFGS-B iterations.



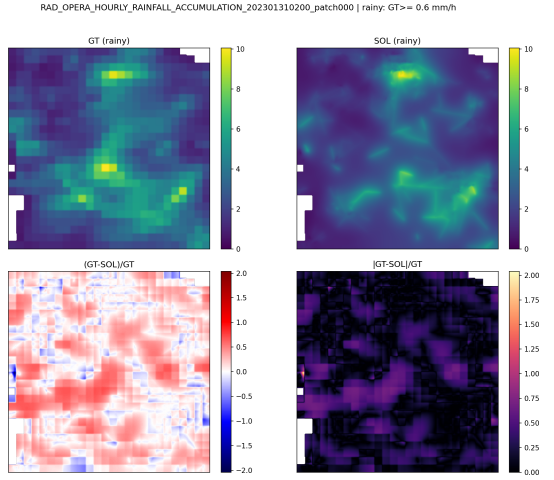
(a) Map of the area of the rain event



(b) IDW



(c) ILDW



(d) Nonlinear optimizer

Figure 8: Case study for a rain event registered on January 31st, 2023. The left panel provides the map view of the selected patch, while the three solver panels show, for IDW, ILDW, and the nonlinear optimizer, the rainy-pixel ground truth field R_P , the estimate $\hat{R}_{P,s}$, the signed relative error $(R_P - \hat{R}_{P,s})/R_P$, and the absolute relative error $|R_P - \hat{R}_{P,s}|/R_P$. The common layout makes it easier to compare smoothing, displacement, and error localization across methods for the same event.

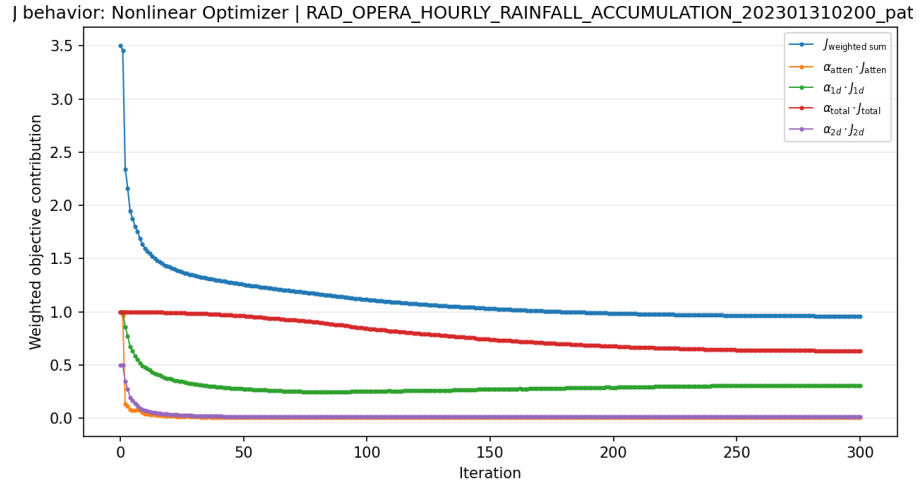
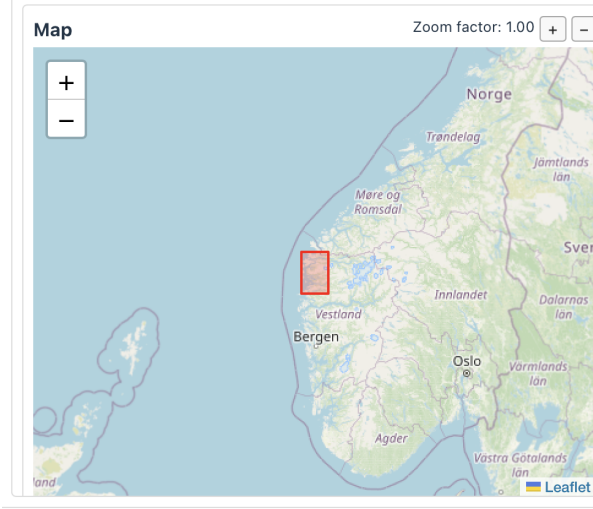
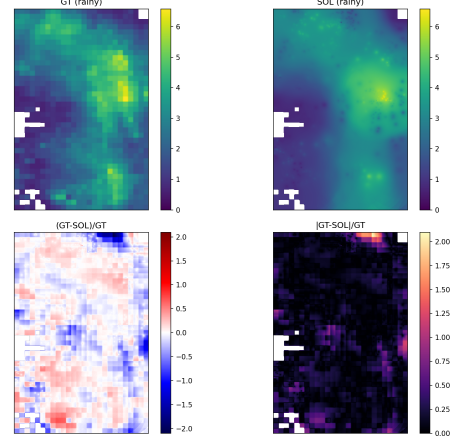


Figure 9: Optimizer trace for the January 31st, 2023 case study at 02:00. As above, the plot reports the iteration-by-iteration decrease of the weighted objective and its weighted constituent terms.



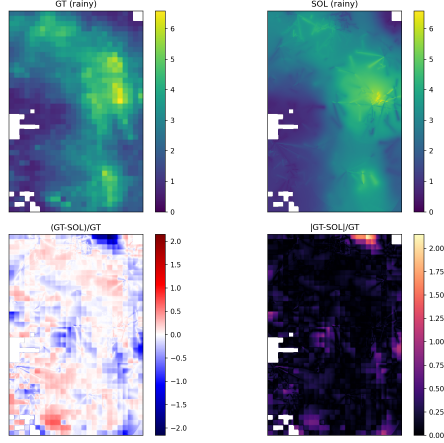
(a) Map of the area of the rain event

RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301311600_patch000 | rainy: GT>= 0.6 mm/h



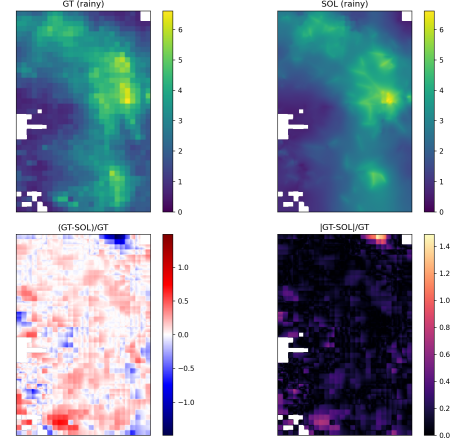
(b) IDW

RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301311600_patch000 | rainy: GT>= 0.6 mm/h



(c) ILDW

RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_202301311600_patch000 | rainy: GT>= 0.6 mm/h



(d) Nonlinear optimizer

Figure 10: Case study for a rain event registered on January 31st, 2023. The map on the left identifies the patch location and footprint, and the three solver panels show the same rainy-pixel diagnostic decomposition as above for IDW, ILDW, and the nonlinear optimizer: R_P , $\hat{R}_{P,s}$, $(R_P - \hat{R}_{P,s})/R_P$, and $|R_P - \hat{R}_{P,s}|/R_P$. Together with the previous two figures, this case gives a compact qualitative comparison of solver behavior across distinct rainfall scenes.

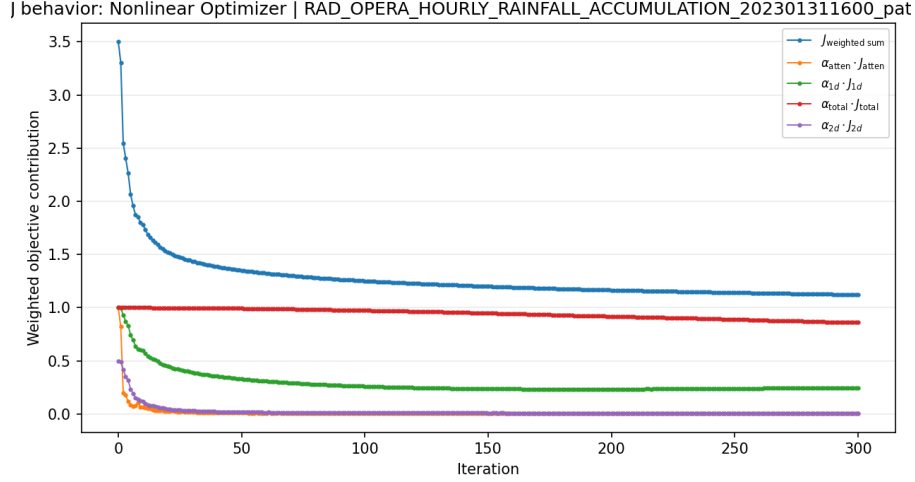


Figure 11: Optimizer trace for the January 31st, 2023 case study at 16:00. The weighted objective and weighted component terms are shown over the L-BFGS-B iterations for the same patch displayed in the spatial case-study panels above.

Optimization Algorithm

The nonlinear optimizer used in this note is L-BFGS-B, a limited-memory quasi-Newton method for smooth constrained optimization. In the present setting, the optimization variable is the rainfall value at every pixel of the patch, so the problem dimension is the number of pixels in the patch. Rather than storing a full Hessian matrix, L-BFGS-B builds a low-memory approximation to local curvature information from the most recent iterates and gradients, and uses that approximation to propose a descent direction. A line search is then carried out along that direction, and the process is repeated until the stopping criteria on iteration count, function reduction, or projected gradient norm are met.

The suffix “-B” indicates that the method handles box constraints. Here those constraints enforce nonnegativity of the rainfall field, so every pixel estimate is required to remain in $[0, \infty)$. In practice, this means that the optimizer can exploit gradient-based curvature information while still respecting the physically natural requirement that rainfall rates should not become negative. The optimizer is initialized from the ILDW estimate, and the cost-reduction plots in the previous section illustrate how the weighted objective decreases from that starting point over successive iterations.

IDW and ILDW Pseudocode

For reference, the no-comments pseudocode for the IDW and ILDW baselines is included below.

Pseudocode for IDW and ILDW

Notation. Each link ℓ is the segment $[p_{\ell,1}, p_{\ell,2}]$ with endpoints $p_{\ell,1}, p_{\ell,2} \in \mathbb{R}^2$, midpoint $m_\ell = (p_{\ell,1} + p_{\ell,2})/2$, and length $L_\ell = \|p_{\ell,2} - p_{\ell,1}\|_2$. Each pixel p has center $c_p \in \mathbb{R}^2$.

IDW

Algorithm 1 Discretized Midpoint IDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_\ell, f_\ell, \text{pol}_\ell)$, radius $r_{\max}$

```

1: for each link  $\ell$  do
2:    $L_\ell \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$ 
3:    $(\kappa_\ell, \alpha_\ell) \leftarrow \text{ITU838}(f_\ell, \text{pol}_\ell)$ 
4:    $R_\ell \leftarrow \left( \frac{A_\ell/L_\ell}{\kappa_\ell} \right)^{1/\alpha_\ell}$ 
5: end for
6: for each pixel  $p$  with center  $c_p$  do
7:   if one or more link midpoints lie inside pixel  $p$  then
8:      $\hat{R}_p \leftarrow$  mean of  $R_\ell$  of those links
9:   else
10:     $N_p \leftarrow \{\ell : \|m_\ell - c_p\|_2 \leq r_{\max}\}$ 
11:    if  $N_p = \emptyset$  then
12:       $\hat{R}_p \leftarrow 0$ 
13:    else
14:       $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\|m_\ell - c_p\|_2^{-2}}{\sum_{j \in N_p} \|m_j - c_p\|_2^{-2}} R_\ell$ 
15:    end if
16:  end if
17: end for
18: Return field  $\hat{R}$ 

```

ILDW

Algorithm 2 Segment-Distance ILDW

Require: links $(p_{\ell,1}, p_{\ell,2}, A_{\ell}, f_{\ell}, \text{pol}_{\ell})$, radius $r_{\max}$

```

1: for each link  $\ell$  do
2:    $L_{\ell} \leftarrow \|p_{\ell,2} - p_{\ell,1}\|_2$ 
3:    $(\kappa_{\ell}, \alpha_{\ell}) \leftarrow \text{ITU838}(f_{\ell}, \text{pol}_{\ell})$ 
4:    $R_{\ell} \leftarrow \left( \frac{A_{\ell}/L_{\ell}}{\kappa_{\ell}} \right)^{1/\alpha_{\ell}}$ 
5: end for
6: for each pixel  $p$  with center  $c_p$  do
7:   if one or more links cross pixel  $p$  then
8:      $\hat{R}_p \leftarrow$  mean of  $R_{\ell}$  of those links
9:   else
10:     $N_p \leftarrow \{\ell : \text{dist}(c_p, [p_{\ell,1}, p_{\ell,2}]) \leq r_{\max}\}$ 
11:    if  $N_p = \emptyset$  then
12:       $\hat{R}_p \leftarrow 0$ 
13:    else
14:       $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\text{dist}(c_p, [p_{\ell,1}, p_{\ell,2}])^{-2}}{\sum_{j \in N_p} \text{dist}(c_p, [p_{j,1}, p_{j,2}])^{-2}} R_{\ell}$ 
15:    end if
16:  end if
17: end for
18: Return field  $\hat{R}$ 

```
